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Grade 11 Geography

4. Massive Igneous Intrusions

Learning Guide

1. What Are Igneous Intrusions?

- **Igneous intrusions** form when **magma enters cracks or spaces inside the Earth's crust** and cools **underground**.
- Because the magma cools slowly underground, it forms **large crystals**.
- Example: **Granite** forms when magma cools slowly below the surface.
- These intrusions are **massive**, meaning they are large bodies of rock and are **not layered** like sedimentary rocks.
- Over time, **weathering and erosion** remove the overlying rock, exposing the intrusion at the surface.
- These features are collectively called **plutonic features**.

📌 **Key idea:**
Igneous intrusions form underground, but later become visible when erosion removes the rocks above them.

2. Overview of Intrusion Types

Intrusion Type	Shape	Typical Rock	Feature Formed
Batholith	Deep and massive	Granite	Granite domes
Laccolith	Mushroom-shaped	Granite/Dolerite	Cuesta domes
Lopolith	Saucer-shaped	Dolerite	Cuesta basins
Sill	Horizontal sheet	Dolerite	Karoo landscape caps
Dyke	Vertical wall	Dolerite	Hogsback-like ridges
Volcanic Pipe	Cylindrical	Dolerite	Magma channel to volcano

📌 **Study tip:**
The shape of the intrusion helps determine the type of landform that may form after erosion.

3. Batholith

- A **batholith** is the **deepest and largest igneous intrusion**.
- It forms about **5–30 km below the surface**.
- It is usually made of **granite**.
- Batholiths create **steep and irregular landforms** when exposed.
- When erosion removes the overlying rock, the exposed batholith may become a **granite dome**.
- Example: **Paarl Mountain**.

📌 **Key idea:**
Batholiths are huge underground granite bodies that can later form granite domes when exposed.

4. Laccolith

- A **laccolith** forms when magma intrudes **between sedimentary rock layers**.
- The magma pushes the rock layers **upward**, forming a **dome shape**.
- It has a **mushroom shape**.
- Laccoliths are found **closer to the surface** than batholiths.

- After erosion, they may lead to the formation of **cuesta domes**.

📌 **Key idea:**
Laccolith = magma pushes upward → dome shape.

🟩 5. Lopolith

- A **lopolith** is a **saucer-shaped intrusion**.
- It forms when sedimentary layers **sag downward**.
- As magma cools, its volume may shrink, creating a space or void.
- The overlying rocks may then **collapse into the space**.
- This produces a basin-like structure.
- Lopoliths can result in **cuesta basins**.
- Example: **Bushveld Igneous Complex**.

📌 **Key idea:**
Lopolith = layers sag downward → basin shape.

🟧 6. Sills

- A **sill** is a **horizontal intrusion**.
- It forms when magma moves **between sedimentary rock layers**.
- Sills are commonly made of **dolerite**.
- They are more resistant to erosion than the surrounding sedimentary rocks.
- In Karoo landscapes, dolerite sills often form **flat caps** on hills.

📌 **Key idea:**
Sill = horizontal sheet of hard rock between layers.

🟥 7. Dykes

- A **dyke** is a **vertical intrusion**.
- It cuts **across rock layers**.
- Dykes can be several metres wide.
- They may form **sharp ridges or wall-like features** when exposed.
- Dykes are **impermeable**, meaning water cannot easily pass through them.
- Because of this, they can help with **groundwater storage**.

📌 **Key idea:**
Dyke = vertical wall of igneous rock cutting across layers.

🟪 8. Volcanic Pipes

- A **volcanic pipe** is a **cylindrical channel**.
- It is the pathway through which magma moves **upward**.
- It often becomes the **centre of a volcano**.
- It is commonly made of **dolerite**.
- It is not always visible at the surface unless surrounding rocks have been eroded away.

📌 **Key idea:**
Volcanic pipe = magma channel to the volcano.

⛰️ Landforms Linked to Massive Igneous Intrusions

🟨 9. Granite Domes

- **Granite domes** are **rounded, smooth landforms** made of granite.
- They form from exposed **batholiths or laccoliths**.
- At first, only a small thin sheet of granite is exposed. This is called a **ruware**.
- With continued erosion, more granite becomes exposed, forming a full **granite dome**.
- During cooling, **orthogonal and curved joints** develop.

- **Exfoliation** causes outer rock layers to peel off, giving the dome a smooth surface.
- Example: **Paarl Mountain**.

Formation of Granite Domes

1. Magma intrudes into the crust and forms a **granite batholith or laccolith**.
2. Erosion removes the Earth's surface above it.
3. The top part of the granite body becomes exposed.
4. The thin exposed sheet is called a **ruware**.
5. More erosion exposes a larger mass of granite.
6. The exposed granite protrudes above the surface as a **granite dome**.

📌 **Key idea:**

Granite dome = exposed granite intrusion shaped by erosion and exfoliation.

🟫 10. Tors

- **Tors** are stacked piles of **rounded boulders**, also called **core stones**.
- They are formed from **granite or dolerite**.
- They develop underground through **chemical weathering along joints**.
- Later, erosion removes the overlying material and exposes the boulders.
- The bottom boulders may still be connected to the **bedrock**.

Formation of Tors

1. Water seeps into cracks and joints in granite below the surface.
2. Chemical weathering takes place along the joints.
3. Erosion exposes the granite mass.
4. Weathered rock is removed by **water and wind**.
5. Exfoliation causes rock layers to peel off.
6. A pile of rounded core stones remains.

📌 **Key idea:**

Tors form when underground weathered granite is exposed and rounded boulders remain.

🧠 11. Final Exam Summary

- **Igneous intrusions** form when magma cools underground.
- Slow cooling forms **large crystals**, such as in granite.
- Intrusions are later exposed by **weathering and erosion**.
- **Batholiths** are deep, massive granite intrusions that can form **granite domes**.
- **Laccoliths** push layers upward and may form **cuesta domes**.
- **Lopoliths** sag downward and may form **cuesta basins**.
- **Sills** are horizontal intrusions and form resistant caps in Karoo landscapes.
- **Dykes** are vertical intrusions that cut across layers and may form ridges.
- **Volcanic pipes** are magma channels leading to volcanoes.
- **Granite domes** are rounded, smooth exposed granite landforms.
- **Tors** are piles of rounded boulders formed by chemical weathering and erosion.
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